Collab -Nanjing Ziye Jia

Target Topics - Drone 2 Drone communication /

ADS-B - ADS-LTE based ..

Paper: Joint ADS-B in 5G

ADS-B is for broacast flight information -1090 Mhz

impact of packet collision and loss

5g is used with Hierarchical Aerial network - MEC

ADS-B , 5G based.
ADS-L proposed by EASA

5G and hirarchical framework -

Geo zone base Air to Air channel

Latency and congestion

Key Points and Gaps from the Two Papers

1. "Joint ADS-B in 5G for Hierarchical Aerial Networks: Performance Analysis and Optimization"

(Source: arXiv)

- **Key Contributions:**
    - Proposes a hierarchical aerial network integrating **ADS-B and 5G** to enhance UAV communication and surveillance.
    - Develops **Air-to-Ground (A2G) and Air-to-Air (A2A) channel models** using stochastic geometry.
    - Introduces a **multi-layer UAV structure**, with **central UAVs** coordinating lower-tier UAVs.
    - Derives signal-to-interference-plus-noise ratio (**SINR**) expressions and evaluates **coverage probability**.

    - Implements **on-board processing with edge computing** to handle redundant or missing ADS-B packets.
- **Gaps and Limitations:**
    - **Interference from non-cooperative sources** (e.g., other UAVs, terrestrial networks) is not fully considered.
    - **Real-world validation** using field experiments is missing.
    - **Latency and congestion** in high-density UAV operations need further study.

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2. "Interference Analysis for Coexistence of UAVs and Civil Aircrafts Based on ADS-B"

(Source: IEEE Xplore)

- **Key Contributions:**
    - Investigates **ADS-B signal interference** between UAVs and civil aircraft.
    - Analyzes the impact of **high UAV density** on ADS-B message integrity.
    - Evaluates **signal collision and packet loss rates** in shared airspace.
    - Recommends **interference mitigation strategies**, such as frequency filtering and adaptive transmission.
- **Gaps and Limitations:**
    - Lacks integration with **5G networks**, unlike the first paper.
    - Does not propose an alternative **communication protocol** for UAVs to reduce ADS-B congestion.
    - Does not account for **multi-tier UAV networks**, which may improve spectral efficiency.

Comparison and Research Gaps

- **Overlap:** Both papers address **UAV communication challenges in shared airspace** and highlight interference concerns.
- **Gaps in Research:**

- A unified **5G-ADS-B framework** to optimize UAV and civil aircraft coexistence.
- **Advanced interference mitigation** techniques beyond frequency filtering.
- **Scalability studies** for high-density UAV deployments.
- **Real-world testing** and performance validation.

Yes, a **5G-ADS-B framework** can be integrated into a **Geo-zones-based solution** inspired by **EASA's U-space** concept. This can enhance **latency management and congestion control** in UAV operations. Here's how:

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1. Geo-zones & 5G-ADS-B Integration

The **European Union Aviation Safety Agency (EASA)** introduced **U-space Geo-zones**, which define controlled airspace areas with specific rules for UAV operations. These zones can be leveraged for **adaptive 5G-ADS-B management**:

- **Strategic Deconfliction:** Assign UAVs to dynamic **5G spectrum slots** based on real-time congestion levels.
- **Priority Access:** Emergency UAVs (e.g., medical drones) can be given **higher transmission priority**.
- **Geo-fencing & Network Slicing:** 5G enables **network slicing** to allocate bandwidth dynamically within different Geo-zones.

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2. Addressing Latency and Congestion

A **collaborative 5G-ADS-B solution** using Geo-zones can help mitigate:

- **ADS-B Congestion:** UAVs in high-traffic zones can switch between **5G-V2X (Vehicle-to-Everything)** and ADS-B to prevent data overload.
- **Low-latency Communication:** Edge computing within Geo-zones can process UAV data locally, reducing ADS-B dependency.
- **Adaptive Transmission:** UAVs in congested areas can **switch transmission rates** based on real-time interference levels.

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3. Possible Collaboration Models

- **Regulators (EASA, FAA):** Define **U-space Geo-zones** for UAV communication standards.
- **Telecom Providers:** Implement **5G network slicing** for UAV priority access.
- **Air Navigation Service Providers (ANSPs):** Ensure **real-time airspace management**.
- **Industry (UAV Operators & Manufacturers):** Develop **hybrid ADS-B & 5G-enabled UAVs**.

Would you like a more detailed breakdown of technical implementation strategies? 🚀

Technical Implementation Strategies for Strategic Deconfliction, Geo-fencing & Network Slicing in a 5G-ADS-B Framework

A **5G-ADS-B hybrid framework** can significantly improve UAV operations within **EASA's U-space Geo-zones** by ensuring safe, low-latency, and congestion-free aerial traffic management. Here's how **Strategic Deconfliction, Geo-fencing, and Network Slicing** can be implemented:

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1. Strategic Deconfliction

Strategic deconfliction involves **preemptive UAV trajectory management** to minimize the risk of mid-air collisions and reduce congestion.

Implementation Strategies

🔷 **Dynamic Flight Path Allocation (AI-Driven Conflict Resolution)**

- Implement **AI-based trajectory prediction models** (e.g., **Deep Reinforcement Learning**) to optimize UAV flight paths dynamically.
- Use **stochastic geometry** models to predict interference hotspots in the **Air-to-Ground (A2G)** and **Air-to-Air (A2A)** spectrum.
- Prioritize UAV traffic in **high-density areas** using **5G Multi-access Edge Computing (MEC)** to process real-time ADS-B and 5G-V2X (Vehicle-to-Everything) data.

🔷 **Time-Sensitive Networking (TSN) for UAV Traffic Management**

- Apply **TSN protocols over 5G** to prioritize UAV communication **based on urgency (e.g., emergency UAVs vs. delivery drones).**
- Use **IEEE 802.1Qbv (Time-Aware Shaper)** to schedule transmissions and avoid packet collision.

🔷 **Decentralized Blockchain-Based Flight Authorization**

- **Smart contracts** on a blockchain network can **automatically validate UAV flight plans** within each **Geo-zone**, preventing unauthorized entries.
- This ensures **secure, conflict-free airspace management** while reducing ADS-B congestion.

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2. Geo-fencing

Geo-fencing creates **virtual boundaries in airspace** that **restrict or regulate UAV movement** using GPS, ADS-B, and 5G.

Implementation Strategies

🔷 **Cloud-based U-space Geo-fencing System**

- Develop a **cloud-based Geo-zone database** that integrates **ADS-B, 5G location services, and UTM (UAV Traffic Management) systems**.
- UAVs continuously **cross-check real-time airspace permissions** via 5G-connected **edge nodes**.

🔷 **Adaptive Geo-fencing via Network Slicing & AI**

- Implement **adaptive Geo-fencing**, where UAVs receive dynamic updates **based on network congestion, weather, and emergency events**.
- AI-based algorithms adjust **fencing parameters in real-time**, ensuring safe UAV rerouting.

🔷 **Automated Geo-zone Violation Alerts & Countermeasures**

- UAVs that enter **restricted zones (e.g., airports, no-fly zones)** trigger **automated alerts**.

- Countermeasures like **RF jamming, UAV interception, or automated UAV landing** are applied **based on threat level**.

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3. 5G Network Slicing for UAV Traffic Optimization

5G network slicing allows UAVs to operate **on separate virtual networks**, ensuring **optimal Quality of Service (QoS)** for different UAV types.

Implementation Strategies

🔷 **Dedicated Slices for UAV Categories**

- **Low-latency slices (URLLC - Ultra-Reliable Low-Latency Communication):**
    - Used for **emergency UAVs (e.g., medical drones, law enforcement).**
    - Guarantees **1-5ms latency** via **5G MEC offloading** for real-time ADS-B updates.
- **Enhanced Mobile Broadband (eMBB) Slices:**
    - Allocated for **delivery UAVs, logistics drones, and passenger eVTOLs.**
- **Massive Machine Type Communications (mMTC) Slices:**
    - Handles **non-time-critical UAV swarm operations** (e.g., environmental monitoring).

🔷 **AI-based Slice Orchestration**

- AI-driven **network slicing managers** monitor UAV density, signal congestion, and interference.
- **Dynamic reallocation of 5G bandwidth** ensures optimal performance **even in high-traffic zones.**

🔷 **Federated Learning for Airspace Adaptation**

- **Distributed AI models** learn from UAV network conditions in different Geo-zones.
- UAVs in congested zones **preemptively switch communication methods** (e.g., **from ADS-B to 5G-V2X**).

🔗 Conclusion: A Unified 5G-ADS-B Geo-zone System

A **strategically deconflicted, geo-fenced, and network-sliced** airspace ensures **scalable, real-time, and collision-free UAV operations.**

This framework is a collaboration between **aviation regulators (EASA, FAA), telecom providers (5G infrastructure), and UAV operators.**

Would you like a **flowchart or system architecture diagram** to visualize the implementation? 🚀

Here's a professional email draft for reaching out to the author, highlighting their research contributions and proposing potential collaboration points:

Subject: Potential Collaboration on 5G-ADS-B Framework & Geo-Zone Integration

Dear [Author's Name],

I hope this email finds you well. I recently had the opportunity to read your insightful work on **"[Title of Paper – Joint ADS-B in 5G for Hierarchical Aerial Networks / Interference Analysis for Coexistence of UAVs and Civil Aircraft]"** and was particularly impressed by your approach to integrating **ADS-B with 5G for UAV operations**. Your research provides a strong foundation for enhancing hierarchical aerial networks and optimizing UAV surveillance.

As we explore next-generation UAV traffic management, I see a strong potential for extending your framework by integrating **Geo-zones-based solutions (inspired by EASA's U-space concept)** to address key challenges like **latency, congestion, and interference mitigation**. Specifically, we see collaboration opportunities in:

1. **Strategic Deconfliction** – Utilizing **AI-based trajectory prediction and TSN (Time-Sensitive Networking) over 5G** to dynamically optimize UAV flight paths and mitigate congestion in high-density areas.
2. **Geo-fencing & Adaptive Routing** – Implementing **real-time cloud-based U-space Geo-fencing** with AI-driven adaptive UAV rerouting to manage interference between UAVs and civil aircraft.
3. **5G Network Slicing for UAV Operations** – Developing a dedicated network slicing model, leveraging **URLLC for critical UAV missions, eMBB for**

commercial UAVs, and mMTC for swarm operations, ensuring efficient spectrum allocation.